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AIR MATTRESS MANUFACTURING APPARATUS, AIR MATTRESS MANUFACTURING METHOD USING SAME, AND AIR MATTRESS MANUFACTURED USING MANUFACTURING METHOD

Abstract

An air mattress manufacturing apparatus according to the present invention comprises: a first support jig having a upper surface on which a center portion of a shape-maintaining member temporarily attached to a corner side surface of a mattress body is placed by a worker; a first pressing unit for pressing the center portion of the shape-maintaining member placed on the upper surface of the first support jig and adhering same to the corner side surface of the mattress body; a second support jig having a upper surface on which the upper end portion of the shape-maintaining member temporarily attached to a corner upper surface of the mattress body and the lower end portion of the shape-maintaining member temporarily attached to a corner lower surface of the mattress body are placed by the worker after the center portion of the shape-maintaining member adhered to the corner side surface of the mattress body by the first pressing unit is vertically folded in half together with the corner side surface of the mattress body; and a second pressing unit for pressing the upper end portion of the shape-maintaining member placed on the upper surface of the second support jig and adhering same to the corner upper surface of the mattress body, or pressing the lower end portion of the shape-maintaining member placed on the upper surface of the second support jig and adhering same to the corner lower surface of the mattress body, and thus the present invention can firmly adhere the shape-maintaining members to four corner portions of the mattress body, and can prevent burns to the worker.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to an air mattress manufacturing apparatus, an air mattress manufacturing method using the same, and an air mattress manufactured using the manufacturing method, and in particular, present invention relates to an air mattress manufacturing apparatus for adhering shape-maintaining members to four corner portions of a mattress body so that the mattress body maintains a hexahedron, an air mattress manufacturing method using the same, and an air mattress manufactured using the manufacturing method.

BACKGROUND ART

[0002] Typically, a bed consists of a bed frame and a mattress that is installed on the upper surface of the bed frame to support the user.

[0003] As for the above mattress, a mattress having a plurality of springs installed therein to support the load of a user using cushioning force, or formed of a material having its own cushioning force, such as latex foam, is commonly used, although mattresses having water or air injected therein are also used.

[0004] Among these, air mattresses that are filled with air inside are not only light but also have the advantage of being easy to adjust for strength and height by controlling the amount of air injected inside.

[0005] Recently, a string air mattress has been developed in which the upper plate portion and lower plate portion of the air mattress are connected by a plurality of strings. The string air mattress has the cushioning force of the air filled inside as well as the elasticity of the strings provided inside, so it has the advantage of being light in weight but with greatly enhanced cushioning force.

[0006] Korean Patent Registration No. 10-1980575 (May 21, 2019) (hereinafter referred to as “prior art”) discloses a technology related to the string air mattress, “Method of manufacturing 3D air mat and it made thereby.”

[0007] The above prior art adheres a sealing tape to the entire edge of the interlocked lower fabric and upper fabric of a semi-assembled air mattress, and then adheres a shape-maintaining member from the lower surface of the lower coated surface of the lower fabric to the outer surface of the sealing tape and the upper surface of the upper coated surface of the upper fabric so that the corner portion of the air mattress is not distorted or crumpled.

[0008] However, in the above-mentioned prior art, since the adhesion process of the shape-maintaining member was performed manually by a worker, there was a problem of reduced adhesive strength, and there was also a problem of the worker sustaining burns in the process of applying hot air to the shape-maintaining member to improve the adhesive strength of the shape-maintaining member.

DISCLOSURE

Technical Problem

[0009] The problem to be solved by the present invention is to provide an air mattress manufacturing apparatus capable of firmly adhering a shape-maintaining member to the four corner portions of a mattress body and preventing burns to a worker, an air mattress manufacturing method using the same, and an air mattress manufactured using the manufacturing method.

[0010] The tasks of the present invention are not limited to the tasks mentioned above, and other tasks not mentioned will be clearly understood by those skilled in the art from the description below.

Technical Solution

[0011] In order to achieve the above task, an air mattress manufacturing apparatus according to the present invention is an apparatus for adhering a shape-maintaining member to each of the four corner portions of a mattress body. Here, the shape-maintaining members are adhered to the four corner portions of the mattress body to maintain the mattress body in a hexahedron. The air mattress manufacturing apparatus according to the present invention includes a first support jig, a first pressing unit, a second support jig, and a second pressing unit. A center portion of the shape-maintaining member temporarily attached to a corner side surface of the mattress body is placed on an upper surface of the first support jig by a worker. The first pressing unit presses the center portion of the shape-maintaining member placed on the upper surface of the first support jig to adhere it to the corner side surface of the mattress body. After the center portion of the shape-maintaining member adhered to the corner side surface of the mattress body by the first pressing unit is vertically folded in half together with the corner side surface of the mattress body by the worker, an upper end portion of the shape-maintaining member temporarily attached to a corner upper surface of the mattress body and a lower end portion of the shape-maintaining member temporarily attached to a corner lower surface of the mattress body are placed on an upper surface of the second support jig by the worker. The second pressing unit presses the upper end portion of the shape-maintaining member placed on the upper surface of the second support jig to adhere it to the corner upper surface of the mattress body, or presses the lower end portion of the shape-maintaining member placed on the upper surface of the second support jig to adhere it to the corner lower surface of the mattress body.

[0012] The upper surface of the first support jig may be formed as a horizontal plane. The upper surface of the second support jig may be formed as an upwardly pointed cone shape.

[0013] The first pressing unit may include a first heating presser and a first cooling presser. The first heating presser may heat and press the center portion of the shape-maintaining member placed on the upper surface of the first support jig. The first cooling presser may press the center portion of the shape-maintaining member pressed by the first heating presser without applying heat. The second pressing unit may include a second heating presser and a second cooling presser. The second heating presser may heat and press the upper end portion of the shape-maintaining member placed on the upper surface of the second support jig, or the lower end portion of the shape-maintaining member placed on the upper surface of the second support jig. The second cooling presser may press the upper end portion of the shape-maintaining member pressed by the second heating presser without applying heat, or may press the lower end portion of the shape-maintaining member pressed by the second heating presser without applying heat.

[0014] The first pressing unit may further include a first guide jig. The first guide jig may have a first guide hole formed therein. The first guide jig may be moved to the first support jig such that an upper end portion of the first support jig is inserted into the first guide hole, thereby fixing the mattress body placed on the upper surface of the first support jig. The second pressing unit may further include a second guide jig. The second guide jig may have a second guide hole formed therein. The second guide jig may be moved to the second support jig such that an upper end portion of the second support jig is inserted into the second guide hole, thereby fixing the mattress body placed on the upper surface of the second support jig.
[0015] The first guide jig and the second guide jig may be arranged to be movable forward, backward, upward and downward.
[0016] The first pressing unit may further include a first linear rail and a first alignment slider. The first alignment slider may have the first heating presser and the first cooling presser installed thereon. The first alignment slider may be slid leftward and rightward along the first linear rail to align one of the first heating presser and the first cooling presser to an upper side of the first guide hole. The second pressing unit may further include a second linear rail and a second alignment slider. The second alignment slider may have the second heating presser and the second cooling presser installed thereon. The second alignment slider may be slid leftward and rightward along the second linear rail to align one of the second heating presser and the second cooling presser to an upper side of the second guide hole.
[0017] An air mattress manufacturing method according to the present invention is a method for adhering a shape-maintaining member to each of the four corner portions of a mattress body. Here, the shape-maintaining members are adhered to the four corner portions of the mattress body to maintain the mattress body in a hexahedron. The air mattress manufacturing method according to the present invention includes a first preparation step, a first adhesion step, a second preparation step, and a second adhesion step. In the first preparation step, the center portion of the shape-maintaining member temporarily attached to the corner side surface of the mattress body is placed on the upper surface of a first support jig by a worker. In the first adhesion step, the center portion of the shape-maintaining member placed on the upper surface of the first support jig is pressed by a first pressing unit to be adhered to the corner side surface of the mattress body. In the second preparation step, the center portion of the shape-maintaining member adhered to the corner side surface of the mattress body by the first pressing unit is vertically folded in half together with the corner side surface of the mattress body by the worker, and an upper end portion of the shape-maintaining member temporarily attached to the corner upper surface of the mattress body and a lower end portion of the shape-maintaining member temporarily attached to the corner lower surface of the mattress body on the upper surface of the second support jig by the worker. In the second adhesion step, the upper end portion of the shape-maintaining member placed on the upper surface of the second support jig is pressed by the second pressing unit to be adhered to the corner upper surface of the mattress body, or the lower end portion of the shape-maintaining member placed on the upper surface of the second support jig is pressed by the second pressing unit to be adhered to the corner lower surface of the mattress body.
[0018] The upper surface of the first support jig may be formed as a horizontal plane. The upper surface of the second support jig may be formed as an upwardly pointed cone shape.
[0019] The first adhesion step may include a first-first adhesion step and a first-second adhesion step. In the first-first adhesion step, the center portion of the shape-maintaining member placed on the upper surface of the first support jig may be heated and pressed by a first heating presser of the first pressing unit. In the first-second adhesion step, the center portion of the shape-maintaining member pressed by the first heating presser may be pressed by a first cooling presser of the first pressing unit without applying heat. The second adhesion step may include a second-first adhesion step and a second-second adhesion step. In the second-first adhesion step, the upper end portion of the shape-maintaining member placed on the upper surface of the second support jig may be heated

and pressed by a second heating presser of the second pressing unit, or the lower end portion of the shape-maintaining member placed on the upper surface of the second support jig may be heated and pressed by the second heating presser. In the second-second adhesion step, the upper end portion of the shape-maintaining member pressed by the second heating presser may be pressed by a second cooling presser of the second pressing unit without applying heat, or the lower end portion of the shape-maintaining member pressed by the second heating presser may be pressed by the second cooling presser without applying heat.

[0020] The air mattress manufacturing method according to the present invention may further include a first mattress body fixing step and a second mattress body fixing step. The first mattress body fixing step may be performed after the first preparation step and before the first-first adhesion step. In the first mattress body fixing step, the mattress body placed on the upper surface of the first support jig may be fixed by moving a first guide jig of the first pressing unit such that the upper end portion of the first support jig is inserted into a first guide hole formed in the first guide jig. The second mattress body fixing step may be performed after the second preparation step and before the second-first adhesion step. In the second mattress body fixing step, the mattress body placed on the upper surface of the second support jig may be fixed by moving a second guide jig of the second pressing unit such that the upper end portion of the second support jig is inserted into a second guide hole formed in the second guide jig.

[0021] The first guide jig and the second guide jig may be arranged to be movable forward, backward, upward and downward.

[0022] The air mattress manufacturing method according to the present invention may further include a first alignment step, a second alignment step, a third alignment step, and a fourth alignment step. The first alignment step may be performed after the first mattress body fixing step and before the first-first adhesion step. In the first alignment step, a first alignment slider of the first pressing unit, on which the first heating presser and the first cooling presser are installed, may be slid leftward and rightward along a first linear rail of the first pressing unit to align the first heating presser to an upper side of the first guide hole. The second alignment step may be performed after the first-first adhesion step and before the first-second adhesion step. In the second alignment step, the first alignment slider may be slid leftward and rightward along the first linear rail to align the first cooling presser to the upper side of the first guide hole. The third alignment step may be performed after the second mattress body fixing step and before the second-first adhesion step. In the third alignment step, a second alignment slider of the second pressing unit, on which the second heating presser and the second cooling presser are installed, may be slid leftward and rightward along a second linear rail of the second pressing unit to align the second heating presser to an upper side of the second guide hole. The fourth alignment step may be performed after the second-first adhesion step and before the second-second adhesion step. In the fourth alignment step, the second alignment slider may be slid leftward and rightward along the second linear rail to align the second cooling presser to the upper side of the second guide hole.

[0023] An air mattress according to the present invention is manufactured using an air mattress manufacturing method according to the present invention.

[0024] Specific details of other embodiments are included in the detailed description and drawings.

Advantageous Effects

[0025] An air mattress manufacturing apparatus, an air mattress manufacturing method using the same, and an air mattress manufactured using the manufacturing method according to the present invention first adhere a center portion of a shape-maintaining member to four corner portions of a mattress body, and then adhere an upper end portion of the shape-maintaining member or a lower end portion of the shape-maintaining member to the four corner portions of the mattress body, and thus the shape-maintaining member can be firmly adhered to the four corner portions of the mattress body, and there is an effect of preventing burns to the worker because the worker does not directly apply hot air to the shape-maintaining member.

[0026] The effects of the present invention are not limited to the effects mentioned above, and other effects not mentioned will be clearly understood by those skilled in the art from the description of the claims.

Description

DESCRIPTION OF DRAWINGS

[0027] FIG. **1** is a drawing illustrating an air mattress manufactured by an air mattress manufacturing apparatus according to an embodiment of the present invention,

[0028] FIG. **2** is a cross-sectional view of an edge portion of the air mattress illustrated in FIG. **1**,

[0029] FIG. **3** is a perspective view illustrating an air mattress manufacturing apparatus according to an embodiment of the present invention,

[0030] FIG. **4** is a front view of the first pressing unit and the second pressing unit illustrated in FIG. **3**,

[0031] FIG. **5** is a right side view of the first pressing unit and the second pressing unit illustrated in FIG. **3**,

[0032] FIG. **6** is a drawing illustrating a state in which the upper end portion of the shape-maintaining member is adhered to the corner upper surface of the mattress body by the second pressing unit illustrated in FIG. **3**,

[0033] FIG. **7** is a flow chart illustrating a process of adhering the center portion of the shape-maintaining member to the corner side surface of the mattress body with the first pressing unit illustrated in FIG. **3**, and

[0034] FIG. **8** is a flowchart illustrating a process of adhering the upper end portion of the shape-maintaining member to the corner upper surface of the mattress body or adhering the lower end portion of the shape-maintaining member to the corner lower surface of the mattress body with the second pressing unit illustrated in FIG. **3**.

MODE FOR DISCLOSURE

[0035] Hereinafter, an air mattress manufacturing apparatus, an air mattress manufacturing method using the same, and an air mattress manufactured using the manufacturing method according to an embodiment of the present invention will be described with reference to the drawings.

[0036] FIG. **1** is a drawing illustrating an air mattress manufactured by an air mattress manufacturing apparatus according to an embodiment of the present invention, and FIG. **2** is a cross-sectional view of an edge portion of the air mattress illustrated in FIG. **1**.

[0037] Referring to FIGS. **1** and **2**, an air mattress **10** manufactured by an air mattress manufacturing apparatus **1** (see FIG. **3**) according to an embodiment of the present invention may include a mattress body **11** tailored to fit the size of the air mattress **10** (e.g., king size, queen size, super single size, etc.), a sealing tape **14** adhered to the edge portion of the mattress body **11**, and a shape-maintaining member **15** adhered to the four corner portions of the mattress body **11**.

[0038] The mattress body **11** may be formed as a hexahedron by a shape-maintaining member **15** adhered to the four corner portions of the mattress body **11**, and in this state, the sealing tape **14** may seal the four side surfaces of the mattress body **11**.

[0039] The mattress body **11** may include an upper plate portion **11**A, a lower plate portion **11**B, and a plurality of strings **11**C. However, the technical feature of the present invention is to adhere the shape-maintaining member **15** to the four corner portions of the mattress body **11** constituting the air mattress **10** at an accurate position, and in order to achieve this technical feature, the mattress body **11** may be constituted by only the upper plate portion **11**A and the lower plate portion **11**B without including the plurality of strings **11**C.

[0040] That is, when a plurality of strings **11**C are not provide, the air mattress **10** may be manufactured by sealing the edges of the upper plate portion **11**A and the lower plate portion **11**B with a sealing tape **14** in a state in which the upper plate portion **11**A and the lower plate portion **11**B are overlapped, and then adhering shape-maintaining members **15** to the four corner portions of the mattress body **11**, and the air mattress **10** manufactured in this way can have cushioning force only by the pressure of the air filled inside.

[0041] In addition, even when a plurality of strings **11**C are provide, the air mattress **10** may be manufactured by sealing the edges of the upper plate portion **11**A and the lower plate portion **11**B with a sealing tape **14** in a state in which the upper plate portion **11**A and lower plate portion **11**B are overlapped, and then adhering shape-maintaining members **15** to the four corner portions of the mattress body **11**, and the air mattress **10** manufactured in this way can have more cushioning force due to the elastic force of the plurality of strings **11**C in addition to the pressure of the air filled inside.

[0042] The upper plate portion **11**A may be formed in a rectangular shape, and the lower plate portion **11**B may be formed in a rectangular shape having the same size as the upper plate portion **11**A.

[0043] The upper plate portion **11**A and the lower plate portion **11**B are tailored to fit the size of the air mattress **10** (e.g., king size, queen size, super single size, etc.), and then overlapped with each other in a vertical direction, and the edges of the mattress body **11** are sealed with a sealing tape **14** to form the basic shape of the air mattress **10**.

[0044] As such, after forming the basic shape of the air mattress **10**, the shape-maintaining members **15** may be adhered to the four corner portions of the mattress body **11** from the upper surface of the upper plate portion **11**A to the lower surface of the lower plate portion **11**B so that the mattress body **11** has a 3D hexahedral shape, and an air injection valve **16** may be installed on one side surface to complete the air mattress **10**.

[0045] The shape-maintaining member **15** may include a center portion **15**A, an upper end portion **15**B, and a lower end portion **15**C. The upper end portion **15**B and the lower end portion **15**C may have the same width, and the center portion **15**A may have a width wider than those of the upper end portion **15**B and the lower end portion **15**C. The center portion **15**A may be adhered to a corner side surface of the mattress body **11**, the upper end portion **15**B may be adhered to a corner upper surface of the mattress body **11**, and the lower end portion **15**C may be adhered to a corner lower surface of the mattress body **11**.

[0046] Particularly, in the present embodiment, the shape-maintaining members **15** may be adhered to the exact adhesion positions in the four corner portions of the mattress body **11** because the center portion **15**A is adhered to the corner side surface of the mattress body **11** before the upper end portion **15**B and the lower end portion **15**C, and then the upper end portion **15**B is adhered to the corner upper surface of the mattress body **11** or the lower end portion **15**C is adhered to the corner lower surface of the mattress body **11**.

[0047] FIG. **3** is a perspective view illustrating an air mattress manufacturing apparatus according to an embodiment of the present invention, FIG. **4** is a front view of the first pressing unit and the second pressing unit illustrated in FIG. **3**, FIG. **5** is a right side view of the first pressing unit and the second pressing unit illustrated in FIG. **3**, and FIG. **6** is a drawing illustrating a state in which the upper end portion of the shape-maintaining member is adhered to the corner upper surface of the mattress body by the second pressing unit illustrated in FIG. **3**.

[0048] Hereinafter, in the description, the directions such as forward, backward, upward, downward, leftward, rightward, and in related terms, are the same directions as those shown in FIG. **3**.

[0049] Referring to FIGS. **3** to **6**, an air mattress manufacturing apparatus **1** according to an embodiment of the present invention may be an apparatus for adhering shape-maintaining members **15** to four corner portions of a mattress body **11**. An adhesive may be applied in advance to one surface of the shape-maintaining member **15** to be adhered to the corner portions of the mattress body **11**.

[0050] With the sealing tape **14** adhered to the edge portions of the upper plate portion **11**A and the lower plate portion **11**B of the mattress body **11** overlapped vertically, the air mattress manufacturing apparatus **1** according to an embodiment of the present invention can maintain the mattress body **11** in a hexahedron by adhering shape-maintaining members **15** to four corner portions of the mattress body **11**. That is, the shape-maintaining member **15** may be adhered to the four corner portions of the mattress body **11** to maintain the mattress body **11** in a hexahedron.
[0051] The air mattress manufacturing apparatus **1** according to an embodiment of the present invention may include a first pressing unit **100**, a second pressing unit **200**, a first support jig **310**, and a second support jig **320**.
[0052] The first pressing unit **100** may be an apparatus for adhering the center portion **15**A of the shape-maintaining member **15** to the corner side surface of the mattress body **11**, and the second pressing unit **200** may be an apparatus for adhering the upper end portion **15**B of the shape-maintaining member **15** to the corner upper surface of the mattress body **11**, or adhering the lower end portion **15**C of the shape-maintaining member **15** to the corner lower surface of the mattress body **11**. The first pressing unit **100** and the second pressing unit **200** may be formed with the same configuration.
[0053] The first support jig **310** and the second support jig **320** may be formed with different shapes in upper surfaces. That is, the upper surface of the first support jig **310** may be formed as a horizontal plane, and the upper surface of the second support jig **320** may be formed as an upwardly pointed cone shape. Specifically, the first support jig **310** may be formed as a cylinder shape with the upper surface formed as a horizontal plane, and the second support jig **320** may be formed as a cylinder shape with the upper surface formed as an upwardly pointed cone shape.
[0054] Of the corner portion of the mattress body **11**, the portion to which the upper end portion or lower end portion of the shape-maintaining member **15** is adhered is formed in a pointed shape at the apex portion where three surfaces meet, and thus, in order to completely support the apex portion, the upper surface of the second support jig **320** may be formed in an upwardly pointed cone shape.
[0055] The center portion **15**A of the shape-maintaining member **15** that is temporarily attached to the corner side surface of the mattress body **11** may be placed on the upper surface of the first support jig **310** by the worker.
[0056] That is, the worker may place the shape-maintaining member **15** temporarily attached to the corner side surface of the mattress body **11** on the upper surface of the first support jig **310** after the center portion **15**A of the shape-maintaining member **15** is temporarily attached to the corner side surface of the mattress body **11**.
[0057] The first pressing unit **100** may press the center portion **15**A of the shape-maintaining member **15** placed on the upper surface of the first support jig **310** to adhere it to the corner side surface of the mattress body **11**.
[0058] That is, the worker may press the center portion **15**A of the shape-maintaining member **15** placed on the upper surface of the first support jig **310** with the first pressing unit **100** to adhere it to the corner side surface of the mattress body **11**.
[0059] After the center portion **15**A of the shape-maintaining member **15** adhered to the corner side surface of the mattress body **11** by the first pressing unit **100** is vertically folded in half together with the corner side surface of the mattress body **11** by the worker, the upper end portion **15**B of the shape-maintaining member **15** temporarily attached to the corner upper surface of the mattress body **11** and the lower end portion **15**C of the shape-maintaining member **15** temporarily attached to the corner lower surface of the mattress body **11** may be placed by the worker on the upper surface of the second support jig **320**.
[0060] That is, after adhering the center portion **15**A of the shape-maintaining member **15** to the corner side surface of the mattress body **11** with the first pressing unit **100**, the worker may vertically fold the center portion **15**A of the shape-maintaining member **15** in half together with the

corner side surface of the mattress body **11**. Thereafter, the worker may temporarily attach the upper end portion **15**B of the shape-maintaining member **15** to the corner upper surface of the mattress body **11** and temporarily attach the lower end portion **15**C of the shape-maintaining member **15** to the corner lower surface of the mattress body **11**. Thereafter, the worker may place the upper end portion **15**B of the shape-maintaining member **15** temporarily attached to the corner upper surface of the mattress body **11** and the lower end portion **15**C of the shape-maintaining member **15** temporarily attached to the corner lower surface of the mattress body **11** on the upper surface of the second support jig **320**.

[0061] The second pressing unit **200** may press the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** to adhere it to the corner upper surface of the mattress body **11**, or press the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** to adhere it to the corner lower surface of the mattress body **11**.

[0062] That is, the worker may press the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** to adhere it to the corner upper surface of the mattress body **11**. Alternatively, the worker may press the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** to adhere it to the corner lower surface of the mattress body **11**.

[0063] Specifically, the worker presses the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** to adhere it to the corner upper surface of the mattress body **11**, then flips the corner portion of the mattress body **11** upside down so that the lower end portion **15**C of the shape-maintaining member **15** faces upward, and then presses the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** to adhere it to the corner lower surface of the mattress body **11**.

[0064] The first pressing unit **100** may include a first heating presser **110** and a first cooling presser **120**.

[0065] The first heating presser **110** may heat and press the center portion **15**A of the shape-maintaining member **15** placed on the upper surface of the first support jig **310**.

[0066] The first heating presser **110** may include a cylinder **111** and a rod **112**. The rod **112** may be withdrawn from the inside of the cylinder **111** or introduced into the inside of the cylinder **111** through the lower end of the cylinder **111**. The rod **112** may be moved vertically by hydraulic pressure within the cylinder **111**.

[0067] The cylinder **111** may constitute the upper part of the first heating presser **110**, and the rod **112** may constitute the lower part of the first heating presser **110**. The lower surface of the rod **112** may be a substantial part that applies heat and presses the center portion **15**A of the shape-maintaining member **15** placed on the upper surface of the first support jig **310**. A plurality of bar-shaped heaters that are heated by electricity may be inserted and installed at the lower end portion of the rod **112**.

[0068] Hereinafter, in the description, the lower surface of the first heating presser **110** may mean the lower surface of the rod **112**.

[0069] The first cooling presser **120** may press the center portion **15**A of the shape-maintaining member **15** pressed by the first heating presser **110** without applying heat. That is, the center portion **15**A of the shape-maintaining member **15** may be cooled by the first cooling presser **120** so that adhesion to the corner side surface of the mattress body **11** can be completed.

[0070] The first cooling presser **120** may include a cylinder **121** and a rod **122**. The rod **122** may be withdrawn from the inside of the cylinder **121** or introduced into the inside of the cylinder **121** through the lower end of the cylinder **121**. The rod **122** may be moved vertically by hydraulic pressure within the cylinder **121**.

[0071] The cylinder **121** may constitute the upper part of the first cooling presser **120**, and the rod **122** may constitute the lower part of the first cooling presser **120**. The lower surface of the rod **122** may be a substantial part that presses the center portion **15**A of the shape-maintaining member **15** pressed by the first heating presser **110** without applying heat.
[0072] Hereinafter, in the description, the lower surface of the first cooling presser **120** may mean the lower surface of the rod **122**.
[0073] Since the upper surface of the first support jig **310** is formed as a horizontal plane, the lower surface of the rod **112**, which is the lower surface of the first heating presser **110**, may be formed as a horizontal plane corresponding to the upper surface of the first support jig **310**, and the lower surface of the rod **122**, which is the lower surface of the first cooling presser **120**, may also be formed as a horizontal plane corresponding to the upper surface of the first support jig **310**.
[0074] The second pressing unit **200** may include a second heating presser **210** and a second cooling presser **220**.
[0075] The second heating presser **210** may heat and press the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320**, or heat and press the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320**.
[0076] The second heating presser **210** may include a cylinder **211** and a rod **212**. The rod **212** may be withdrawn from the inside of the cylinder **211** or introduced into the inside of the cylinder **211** through the lower end of the cylinder **211**. The rod **212** may be moved vertically by hydraulic pressure within the cylinder **211**.
[0077] The cylinder **211** may constitute the upper part of the second heating presser **210**, and the rod **212** may constitute the lower part of the second heating presser **210**. The lower surface of the rod **212** may be a substantial part that heats and presses the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320**, or heats and presses the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320**. A plurality of bar-shaped heaters that are heated by electricity may be inserted and installed at the lower end portion of the rod **212**.
[0078] Hereinafter, in the description, the lower surface of the second heating presser **210** may mean the lower surface of the rod **212**.
[0079] The second cooling presser **220** may press the upper end portion **15**B of the shape-maintaining member **15** pressed by the second heating presser **210** without applying heat, or may press the lower end portion **15**C of the shape-maintaining member **15** pressed by the second heating presser **210** without applying heat. That is, the upper end portion **15**B of the shape-maintaining member **15** may be cooled by the second cooling presser **220** so that adhesion to the corner upper surface of the mattress body **11** can be completed, and the lower end portion **15**C of the shape-maintaining member **15** may be cooled by the second cooling presser **220** so that adhesion to the corner lower surface of the mattress body **11** can be completed.
[0080] The second cooling presser **220** may include a cylinder **221** and a rod **222**. The rod **222** may be withdrawn from the inside of the cylinder **221** or introduced into the inside of the cylinder **221** through the lower end of the cylinder **221**. The rod **222** may be moved vertically by hydraulic pressure within the cylinder **221**.
[0081] The cylinder **221** may constitute the upper part of the second cooling presser **220**, and the rod **222** may constitute the lower part of the second cooling presser **220**. The lower surface of the rod **222** may be a substantial part that presses the upper end portion **15**B of the shape-maintaining member **15** pressed by the second heating presser **210** without applying heat, or presses the lower end portion **15**C of the shape-maintaining member **15** pressed by the second heating presser **210** without applying heat.
[0082] Hereinafter, in the description, the lower surface of the second cooling presser **220** may mean the lower surface of the rod **222**.

[0083] Since the upper surface of the second support jig **320** is formed in an upwardly pointed cone shape, the lower surface of the rod **212**, which is the lower surface of the second heating presser **210**, may be formed in a conical groove corresponding to the upper surface of the second support jig **320**, and the lower surface of the rod **222**, which is the lower surface of the second cooling presser **220**, may also be formed in a conical groove corresponding to the upper surface of the second support jig **320**.

[0084] The first pressing unit **100** may further include a first guide jig **130**. The first guide jig **130** may be arranged to be movable forward, backward, upward, and downward. The first guide jig **130** may be moved forward, backward, upward, and downward by the driving force of the electric motors. In a state in which the center portion **15**A of the shape-maintaining member **15** is placed on the upper surface of the first support jig **310**, the first guide jig **130** may be moved forward and then downward by the driving force of the electric motors, thereby fixing the corner portion of the mattress body **11** placed on the upper surface of the first support jig **310**.

[0085] A first guide hole **135** may be formed in the first guide jig **130**. The first guide hole **135** may be formed in a circular shape corresponding to the shape of the outer circumferential surface of the first support jig **310**. The first guide jig **130** may be moved to the first support jig **310** such that the upper end portion of the first support jig **310** may be inserted into the first guide hole **135**, thereby fixing the mattress body **11** placed on the upper surface of the first support jig **310**.

[0086] The second pressing unit **200** may further include a second guide jig **230**. The second guide jig **230** may be arranged to be movable forward, backward, upward, and downward. The second guide jig **230** may be moved forward, backward, upward, and downward by the driving force of the electric motors. In a state in which the upper end portion **15**B and the lower end portion **15**C of the shape-maintaining member **15** are placed on the upper surface of the second support jig **320**, the second guide jig **230** may be moved forward and then downward by the driving force of the electric motors, thereby fixing the corner portion of the mattress body **11** placed on the upper surface of the second support jig **320**.

[0087] A second guide hole **235** may be formed in the second guide jig **230**. The second guide hole **235** may be formed in a circular shape corresponding to the shape of the outer circumferential surface of the second support jig **320**. The second guide jig **230** may be moved to the second support jig **320** such that the upper end portion of the second support jig **320** is inserted into the second guide hole **235**, thereby fixing the mattress body **11** placed on the upper surface of the second support jig **320**.

[0088] The first pressing unit **100** may further include a first linear rail **140** and a first alignment slider **150**.

[0089] The first linear rail **140** may include a pair of first linear rails **140** that are spaced apart from each other in a front-to-back direction and arranged parallel to each other. The pair of first linear rails **140** may be formed to be long leftward and rightward.

[0090] A first heating presser **110** and a first cooling presser **120** may be installed in the first alignment slider **150**. The cylinder **111** of the first heating presser **110** and the cylinder **121** of the first cooling presser **120** may be arranged to protrude upward from the first alignment slider **150**, and the rod **112** of the first heating presser **110** and the rod **122** of the first cooling presser **120** may be arranged to protrude downward from the first alignment slider **150** by penetrating holes formed in the first alignment slider **150**.

[0091] The first alignment slider **150** may be slid leftward and rightward along the first linear rail **140** by the driving force of the electric motor. The first alignment slider **150** may be slid leftward and rightward along the first linear rail **140** to align one of the first heating presser **110** and the first cooling presser **120** to the upper side of the first guide hole **135**. A state such as FIG. **3** is a state in which the first alignment slider **150** is slid rightward along the first linear rail **140**, and may be a state in which the first heating presser **110** is aligned to the upper side of the first guide hole **135**. In a state such as FIG. **3**, the first alignment slider **150** may be slid leftward along the first linear rail

140 to align the first cooling presser **120** to the upper side of the first guide hole **135**.

[0092] The second pressing unit **200** may further include a second linear rail **240** and a second alignment slider **250**.

[0093] The second linear rail **240** may include a pair of second linear rails **240** that are spaced apart from each other in a front-to-back direction and arranged parallel to each other. The pair of second linear rails **240** may be formed to be long leftward and rightward.

[0094] A second heating presser **210** and a second cooling presser **220** may be installed in the second alignment slider **250**. The cylinder **211** of the second heating presser **210** and the cylinder **221** of the second cooling presser **220** may be arranged to protrude upward from the second alignment slider **250**, and the rod **212** of the second heating presser **210** and the rod **222** of the second cooling presser **220** may be arranged to protrude downward from the second alignment slider **250** by penetrating through holes formed in the second alignment slider **250**.

[0095] The second alignment slider **250** may be slid leftward and rightward along the second linear rail **240** by the driving force of the electric motor. The second alignment slider **250** may be slid leftward and rightward along the second linear rail **240** to align one of the second heating presser **210** and the second cooling presser **220** to the upper side of the second guide hole **235**. A state such as FIG. **3** is a state in which the second alignment slider **250** is slid rightward along the second linear rail **240**, and may be a state in which the second heating presser **210** is aligned to the upper side of the second guide hole **235**. In a state such as FIG. **3**, the second alignment slider **250** may be slid leftward along the second linear rail **240** to align the second cooling presser **220** to the upper side of the second guide hole **235**.

[0096] FIG. **7** is a flow chart illustrating a process of adhering the center portion of the shape-maintaining member to the corner side surface of the mattress body with the first pressing unit illustrated in FIG. **3**.

[0097] Referring to FIG. **7**, a method of adhering a center portion **15**A of a shape-maintaining member **15** to a corner side surface of a mattress body **11** may include a first preparation step S**110** and a first adhesion step S**140**, S**160**.

[0098] In the first preparation step S**110**, a worker may place the center portion **15**A of the shape-maintaining member **15** temporarily attached to the corner side surface of the mattress body **11** on the upper surface of the first support jig **310**.

[0099] Specifically, in the first preparation step S**110**, the worker may temporarily attach the center portion **15**A of the shape-maintaining member **15** to the corner side surface of the mattress body **11**, and then place the center portion **15**A of the shape-maintaining member **15** temporarily attached to the corner side surface of the mattress body **11** on the upper surface of the first support jig **310**.

[0100] In the first adhesion step S**140**, S**160**, the center portion **15**A of the shape-maintaining member **15** placed on the upper surface of the first support jig **310** may be adhered to the corner side surface of the mattress body **11** by pressing it with the first pressing unit **100**.

[0101] The first adhesion step S**140**, S**160** may include a first-first adhesion step S**140** and a first-second adhesion step S**160**.

[0102] In the first-first adhesion step S**140**, the center portion **15**A of the shape-maintaining member **15** placed on the upper surface of the first support jig **310** may be heated and pressed by a first heating presser **110** of the first pressing unit **100**. In this way, the adhesive that was previously applied to the adhesion surface of the center portion **15**A of the shape-maintaining member **15** melts due to the heat of the first heating presser **110**, so that the center portion **15**A of the shape-maintaining member **15** may be adhered to the corner side surface of the mattress body **11**.

[0103] In the first-second adhesion step S**160**, the center portion **15**A of the shape-maintaining member **15** pressed by the first heating presser **110** may be pressed by the first cooling presser **120** of the first pressing unit **100** without applying heat. In this way, as the adhesive melted by the heat of the first heating presser **110** is cooled, the center portion **15**A of the shape-maintaining member **15** may be completely adhered to the corner side surface of the mattress body **11**.

[0104] After the first preparation step S**110**, a first mattress body fixing step S**120** may be performed before the first-first adhesion step S**140**.
[0105] In the first mattress body fixing step S**120**, the mattress body **11** placed on the upper surface of the first support jig **310** may be fixed by moving the first guide jig **130** of the first pressing unit **100** such that the upper end portion of the first support jig **310** is inserted into the first guide hole **135** formed in the first guide jig **130**.
[0106] After the first mattress body fixing step S**120**, a first alignment step S**130** may be performed before the first-first adhesion step S**140**.
[0107] In the first alignment step S**130**, the first heating presser **110** may be aligned to the upper side of the first guide hole **135** of the first guide jig **130** by sliding the first alignment slider **150** of the first pressing unit **100**, on which the first heating presser **110** and the first cooling presser **120** are installed, leftward and rightward along the first linear rail **140** of the first pressing unit **100**. Accordingly, in the first-first adhesion step S**140**, the rod **112** of the first heating presser **110** may be moved downward so that the center portion **15**A of the shape-maintaining member **15** can be accurately pressed.
[0108] After the first-first adhesion step S**140**, a second alignment step S**150** may be performed before the first-second adhesion step S**160**.
[0109] In the second alignment step S**150**, the first cooling presser **120** can be aligned to the upper side of the first guide hole **135** of the first guide jig **130** by sliding the first alignment slider **150** leftward and rightward along the first linear rail **140**. Accordingly, in the first-second adhesion step S**160**, the rod **122** of the first cooling presser **120** may be moved downward so that the center portion **15**A of the shape-maintaining member **15** can be accurately pressed.
[0110] FIG. **8** is a flowchart illustrating a process of adhering the upper end portion of the shape-maintaining member to the corner upper surface of the mattress body or adhering the lower end portion of the shape-maintaining member to the corner lower surface of the mattress body with the second pressing unit illustrated in FIG. **3**.
[0111] Referring to FIG. **8**, the method of adhering the upper end portion **15**B of the shape-maintaining member **15** to the corner upper surface of the mattress body **11** or adhering the lower end portion **15**C of the shape-maintaining member **15** to the corner lower surface of the mattress body **11** may include a second preparation step S**210** and a second adhesion step S**240**, S**260**.
[0112] In the second preparation step S**210**, the worker vertically folds the center portion **15**A of the shape-maintaining member **15** adhered to the corner side surface of the mattress body **11** by the first pressing unit **100** in half together with the corner side surface of the mattress body **11**, and then places the upper end portion **15**B of the shape-maintaining member **15** temporarily attached to the corner upper surface of the mattress body **11** and the lower end portion **15**C of the shape-maintaining member **15** temporarily attached to the corner lower surface of the mattress body **11** on the upper surface of the second support jig **320**.
[0113] Specifically, in the second preparation step S**210**, the worker vertically folds the center portion of the shape-maintaining member adhered to the corner side surface of the mattress body by the first pressing unit in half together with the corner side surface of the mattress body **11**, and then temporarily attaches the upper end portion **15**B of the shape-maintaining member **15** to the corner upper surface of the mattress body **11** and the lower end portion **15**C of the shape-maintaining member **15** to the corner lower surface of the mattress body **11**, and then places the upper end portion **15**B of the shape-maintaining member **15** temporarily attached to the corner upper surface of the mattress body **11** and the lower end portion **15**C of the shape-maintaining member **15** temporarily attached to the corner lower surface of the mattress body **11** on the upper surface of the second support jig **320**.
[0114] In the second adhesion step S**240**, S**260**, the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** may be pressed by the second pressing unit **200** to be adhered to the corner upper surface of the mattress body **11**, or the lower

end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** may be pressed by the second pressing unit **200** to be adhered to the corner lower surface of the mattress body **11**.

[0115] In the second adhesion step S**240**, S**260**, the worker may press the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** so as to adhere it to the corner upper surface of the mattress body **11**, then turn the corner portion of the mattress body **11** upside down, and press the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** so as to adhere it to the corner lower surface of the mattress body **11**. Of course, in the second adhesion step S**240**, S**260**, the worker may press the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** so as to adhere it to the corner lower surface of the mattress body **11**, then turn the corner portion of the mattress body **11** upside down, and then press the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** with the second pressing unit **200** so as to adhere it to the corner upper surface of the mattress body **11**.

[0116] The second adhesion step S**240**, S**260** may include a second-first adhesion step S**240** and a second-second adhesion step S**260**.

[0117] In the second-first adhesion step S**240**, the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** may be heated and pressed by the second heating presser **210** of the second pressing unit **200**, or the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** may be heated and pressed by the second heating presser **210**. When the upper end portion **15**B of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** is heated and pressed by the second heating presser **210**, the adhesive previously applied to the adhesion surface of the upper end portion **15**B of the shape-maintaining member **15** melts due to the heat of the second heating presser **210**, so that the upper end portion **15**B of the shape-maintaining member **15** can be adhered to the corner upper surface of the mattress body **11**. Alternatively, when the lower end portion **15**C of the shape-maintaining member **15** placed on the upper surface of the second support jig **320** is heated and pressed by the second heating presser **210**, the adhesive that was previously applied to the adhesion surface of the lower end portion **15**C of the shape-maintaining member **15** melts due to the heat of the second heating presser **210**, so that the lower end portion **15**C of the shape-maintaining member **15** can be adhered to the corner lower surface of the mattress body **11**.

[0118] In the second-second adhesion step S**260**, the upper end portion **15**B of the shape-maintaining member **15** pressed by the second heating presser **210** may be pressed by the second cooling presser **220** of the second pressing unit **200** without applying heat, or the lower end portion **15**C of the shape-maintaining member **15** pressed by the second heating presser **210** may be pressed by the second cooling presser **220** without applying heat. When the upper end portion **15**B of the shape-maintaining member **15** pressed by the second heating presser **210** is pressed by the second cooling presser **220** without applying heat, the upper end portion **15**B of the shape-maintaining member **15** may be completely adhered to the corner upper surface of the mattress body **11** as the adhesive melted by the heat of the second heating presser **210** is cooled. Alternatively, when the lower end portion **15**C of the shape-maintaining member **15** pressed by the second heating presser **210** is pressed by the second cooling presser **220** without applying heat, the lower end portion **15**C of the shape-maintaining member **15** may be completely adhered to the corner lower surface of the mattress body **11** as the adhesive melted by the heat of the second heating presser **210** is cooled.

[0119] After the second preparation step S**210**, a second mattress body fixing step S**220** may be performed before the second-first adhesion step S**240**.

[0120] In the second mattress body fixing step S**220**, the mattress body **11** placed on the upper surface of the second support jig **320** may be fixed by moving a second guide jig **230** of the second pressing unit **200** such that the upper end portion of the second support jig **320** is inserted into the second guide hole **235** formed in the second guide jig **230**.
[0121] After the second mattress body fixing step S**220**, a third alignment step S**230** may be performed before the second-first adhesion step S**240**.
[0122] In the third alignment step S**230**, the second heating presser **210** may be aligned to the upper side of the second guide hole **235** of the second guide jig **230** by sliding the second alignment slider **250** of the second pressing unit **200**, on which the second heating presser **210** and the second cooling presser **220** are installed, leftward and rightward along the second linear rail **240** of the second pressing unit **200**. Accordingly, in the second-first adhesion step S**240**, the rod **212** of the second heating presser **210** may be moved downward to accurately press the upper end portion **15**B of the shape-maintaining member **15** or accurately press the lower end portion **15**C of the shape-maintaining member **15**.
[0123] After the second-first adhesion step S**240**, a fourth alignment step S**250** may be performed before the second-second adhesion step S**260**.
[0124] In the fourth alignment step S**250**, the second cooling presser **220** may be aligned to the upper side of the second guide hole **235** of the second guide jig **230** by sliding the second alignment slider **250** leftward and rightward along the second linear rail **240**. Accordingly, in the second-second adhesion step S**260**, the rod **222** of the second cooling presser **220** may be moved downward to accurately press the upper end portion **15**B of the shape-maintaining member **15** or accurately press the lower end portion **15**C of the shape-maintaining member **15**.
[0125] As described above, the air mattress manufacturing apparatus **1**, the air mattress manufacturing method using the same and the air mattress **10** manufactured using the manufacturing method according to an embodiment of the present invention first adhere the center portion **15**A of the shape-maintaining member **15** to the four corner portions of the mattress body **11**, and then adhere the upper end portion **15**B of the shape-maintaining member **15** or the lower end portion **15**C of the shape-maintaining member **15** to the four corner portions of the mattress body **11**, and thus the shape-maintaining member **15** can be firmly adhered to the four corner portions of the mattress body **11**, and because the worker does not directly apply hot air to the shape-maintaining member **15**, burns to the worker can be prevented.

Claims

1. An air mattress manufacturing apparatus, for adhering shape-maintaining members to four corner portions of a mattress body, the shape-maintaining members being adhered to the corner portions of the mattress body to maintain the mattress body in a hexahedron, the air mattress manufacturing apparatus comprising: a first support jig having an upper surface on which a center portion of the shape-maintaining member, temporarily attached to a corner side surface of the mattress body, is placed by a worker; a first pressing unit configured to press the center portion of the shape-maintaining member, which is placed on the upper surface of the first support jig, to adhere same to the corner side surface of the mattress body; a second support jig having an upper surface on which an upper end portion of the shape-maintaining member, temporarily attached to a corner upper surface of the mattress body, and a lower end portion of the shape-maintaining member, temporarily attached to a corner lower surface of the mattress body, are placed by the worker, after the center portion of the shape-maintaining member, which is adhered to the corner side surface of the mattress body by the first pressing unit, is vertically folded in half together with the corner side surface of the mattress body by the worker; and a second pressing unit configured to press the upper end portion of the shape-maintaining member, which is placed on the upper surface of the second support jig, to adhere same to the corner upper surface of the mattress body, or press the

lower end portion of the shape-maintaining member, which is placed on the upper surface of the second support jig, to adhere same to the corner lower surface of the mattress body.
2. The air mattress manufacturing apparatus of claim 1, wherein the upper surface of the first support jig is formed as a horizontal plane, and the upper surface of the second support jig is formed as an upwardly pointed cone shape.
3. The air mattress manufacturing apparatus of claim 1, wherein the first pressing unit includes: a first heating presser configured to heat and press the center portion of the shape-maintaining member placed on the upper surface of the first support jig; and a first cooling presser configured to press the center portion of the shape-maintaining member pressed by the first heating presser without applying heat, and the second pressing unit includes: a second heating presser configured to heat and press the upper end portion of the shape-maintaining member placed on the upper surface of the second support jig, or heat and press the lower end portion of the shape-maintaining member placed on the upper surface of the second support jig; and a second cooling presser configured to press the upper end portion of the shape-maintaining member pressed by the second heating presser without applying heat, or press the lower end portion of the shape-maintaining member pressed by the second heating presser without applying heat.
4. The air mattress manufacturing apparatus of claim 3, wherein the first pressing unit further includes: a first guide jig in which a first guide hole is formed, and which is moved to the first support jig such that an upper end portion of the first support jig is inserted into the first guide hole, to fix the mattress body placed on the upper surface of the first support jig, and the second pressing unit further includes: a second guide jig in which a second guide hole is formed, and which is moved to the second support jig such that an upper end portion of the second support jig is inserted into the second guide hole, to fix the mattress body placed on the upper surface of the second support jig.
5. The air mattress manufacturing apparatus of claim 4, wherein the first guide jig and the second guide jig are arranged to be movable forward, backward, upward and downward.
6. The air mattress manufacturing apparatus of claim 4, wherein the first pressing unit further includes: a first linear rail; and a first alignment slider on which the first heating presser and the first cooling presser are installed, the first alignment slider sliding leftward and rightward along the first linear rail to align one of the first heating presser and the first cooling presser to an upper side of the first guide hole, and the second pressing unit further includes: a second linear rail; and a second alignment slider on which the second heating presser and the second cooling presser are installed, the second alignment slider sliding leftward and rightward along the second linear rail to align one of the second heating presser and the second cooling presser to an upper side of the second guide hole.
7. An air mattress manufacturing method, for adhering shape-maintaining members to four corner portions of a mattress body, the shape-maintaining members being adhered to the corner portions of the mattress body to maintain the mattress body in a hexahedron, the air mattress manufacturing method comprising: a first preparation step of placing, by a worker, a center portion of the shape-maintaining member, temporarily attached to a corner side surface of the mattress body, on an upper surface of a first support jig; a first adhesion step of adhering the center portion of the shape-maintaining member, which is placed on the upper surface of the first support jig, to the corner side surface of the mattress body, by pressing same with a first pressing unit; a second preparation step of placing, by the worker, an upper end portion of the shape-maintaining member, temporarily attached to a corner upper surface of the mattress body, and a lower end portion of the shape-maintaining member, temporarily attached to a corner lower surface the mattress body, on an upper surface of a second support jig, after vertically folding, by the worker, the center portion of the shape-maintaining member adhered to the corner side surface of the mattress body by the first pressing unit in half together with the corner side surface of the mattress body; and a second adhesion step of adhering the upper end portion of the shape-maintaining member, which is placed

on the upper surface of the second support jig, to the corner upper surface of the mattress body, by pressing same with a second pressing unit, or adhering the lower end portion of the shape-maintaining member, which is placed on the upper surface of the second support jig, to the corner lower surface of the mattress body, by pressing same with the second pressing unit.

8. The air mattress manufacturing method of claim 7, wherein the upper surface of the first support jig is formed as a horizontal plane, and the upper surface of the second support jig is formed as an upwardly pointed cone shape.

9. The air mattress manufacturing method of claim 7, wherein the first adhesion step includes: a first-first adhesion step of heating and pressing the center portion of the shape-maintaining member, which is placed on the upper surface of the first support jig, with a first heating presser of the first pressing unit; and a first-second adhesion step of pressing the center portion of the shape-maintaining member pressed by the first heating presser, with a first cooling presser of the first pressing unit without applying heat, and the second adhesion step includes: a second-first adhesion step of heating and pressing the upper end portion of the shape-maintaining member, which is placed on the upper surface of the second support jig, with a second heating presser of the second pressing unit, or heating and pressing the lower end portion of the shape-maintaining member, which is placed on the upper surface of the second support jig, with the second heating presser; and a second-second adhesion step of pressing the upper end portion of the shape-maintaining member pressed by the second heating presser, with a second cooling presser of the second pressing unit without applying heat, or pressing the lower end portion of the shape-maintaining member pressed by the second heating presser, with the second cooling presser without applying heat.

10. The air mattress manufacturing method of claim 9, further comprising: a first mattress body fixing step of fixing the mattress body placed on the upper surface of the first support jig, by moving a first guide jig of the first pressing unit such that an upper end portion of the first support jig is inserted into a first guide hole formed in the first guide jig, after the first preparation step and before the first-first adhesion step; and a second mattress body fixing step of fixing the mattress body placed on the upper surface of the second support jig, by moving a second guide jig of the second pressing unit such that an upper end portion of the second support jig is inserted into a second guide hole formed in the second guide jig, after the second preparation step and before the second-first adhesion step.

11. The air mattress manufacturing method of claim 10, wherein the first guide jig and the second guide jig are arranged to be movable forward, backward, upward and downward.

12. The air mattress manufacturing method of claim 10, further comprising: a first alignment step of aligning the first heating presser to an upper side of the first guide hole by sliding a first alignment slider of the first pressing unit, on which the first heating presser and the first cooling presser are installed, leftward and rightward along a first linear rail of the first pressing unit, after the first mattress body fixing step and before the first-first adhesion step; a second alignment step of aligning the first cooling presser to the upper side of the first guide hole by sliding the first alignment slider leftward and rightward along the first linear rail, after the first-first adhesion step and before the first-second adhesion step; a third alignment step of aligning the second heating presser to an upper side of the second guide hole by sliding a second alignment slider of the second pressing unit, on which the second heating presser and the second cooling presser are installed, leftward and rightward along a second linear rail of the second pressing unit, after the second mattress body fixing step and before the second-first adhesion step; and a fourth alignment step of aligning the second cooling presser to the upper side of the second guide hole by sliding the second alignment slider leftward and rightward along the second linear rail, after the second-first adhesion step and before the second-second adhesion step.

13. An air mattress manufactured using the air mattress manufacturing method according to claim 7.